



SOLAR LIGHT For A Portable Toilet

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Despite being more expensive than standard permanent outdoor toilets, portable toilets (Fig. 1) have several significant benefits as these are self-contained and can be placed almost anywhere. These days portable toilets are frequently seen at outdoor areas like construction sites,



Fig. 1: Proposed toilet system

farms, camp sites and even street-sides. Solar power provides an ideal solution for applications where lighting is required in portable

toilets that are located away from a mains supply grid, or the installation is expected to be on the site for a brief period.

Here is a circuit of a self-contained solar light for portable toilet lighting applications. The solar light housed within its own enclosure comprises a solar charger, rechargeable battery pack, white LED light source and PIR motion sensor as an occupancy sensor. The occupancy sensor (motion sensor) activates the light source and, after the booth has been vacated (or when there is no valid motion), the lighting switches off to save battery.

Circuit and working

Circuit diagram of the solar light

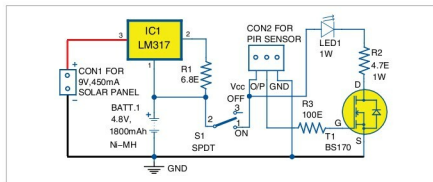


Fig. 2: Circuit diagram of the solar light for a portable toilet

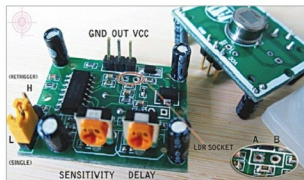


Fig. 3: Photo of the PIR sensor

for a portable toilet is shown in Fig. 2. It is built around a 9V, 450mA solar panel, voltage regulator LM317 (IC1), a PIR sensor, MOSFET BS170 (T1), a 1W white LED (LED1) and a few other components.

The circuit may

PARTS LIST

Semiconductors:

- IC1 - LM317 adjustable voltage regulator with heat-sink
- T1 - BS170 MOSFET
- LED1 - 3.6V, 1-watt white LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 6.8-ohm
- R2 - 4.7-ohm, 1-watt
- R3 - 100-ohm

Miscellaneous:

- S1 - SPDT switch
- CON1 - 2-pin connector terminal
- CON2 - 3-pin connector
- CON3 - 2-pin connector for LED1
- BATT.1 - 4.8V, 1800mAh Ni-MH battery
- 9V, 450mA solar panel
- PIR sensor

be split into two sections: the charger and the sensing and control system. Power is supplied via the 9V, 450mA-rated polycrystalline solar panel, which is linked to CON1 and charges the 4.8V, 1800mAh battery pack, BATT.1, via the constant-current charger built around IC1. The 6.8-ohm resistor (R1) limits the charging current to near one-tenth rate of the battery, that is, 180mA.

The sensing and control circuitry is based on PIR motion detector module (Fig. 3), which is connected across CON2. When a valid motion is detected, output of the PIR sensor goes to around 3.3V and the light source LED1 (3.6V, 1W white LED) is switched on through medium-power MOSFET BS170 for a finite time. The 4.7-ohm, 1W resistor R2 limits the operating current of the white LED.

Construction and testing

An actual-size, single-side PCB for the solar light for a portable toilet is shown in Fig. 4 and its component layout in Fig. 5. Enclose the PCB in